

How healthy is our stock?

On-hand value

£2M

Safety stock value

£679K

Excess value

£371K

Excess % of on-hand

20.3%

Inventory turns

3.2x

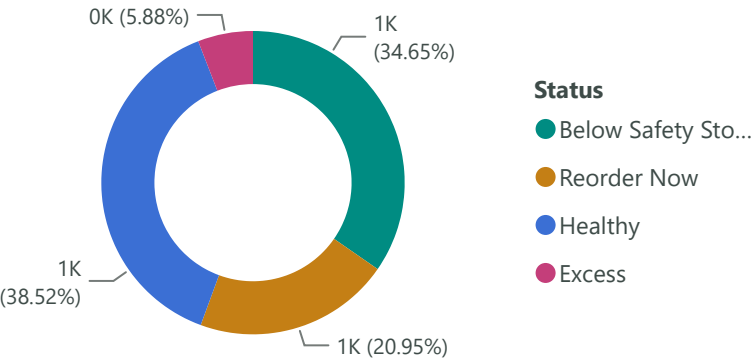
Days on hand

115

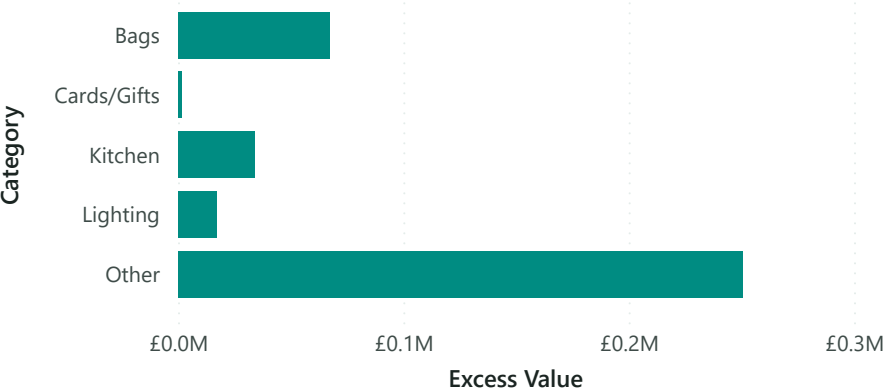
SKUs needing action

2,099

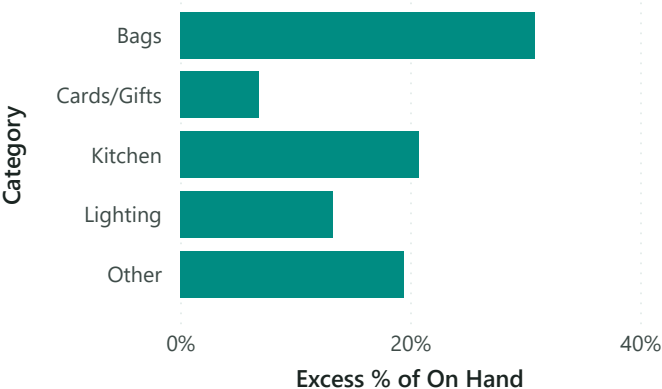
SKU count by stock status



Excess value by category



Excess RATE by category



Category

- ☐ Bags
- ☐ Cards/Gifts
- ☐ Kitchen
- ☐ Lighting

ABC class

- ☐ A
- ☐ B
- ☐ C

Service level mode

- ☐ Class-based (A 98% / B 95%...)
- ☐ Uniform (slider)

A-class below safety stock

229

Annual value at risk

£1M

THE BRIEF - Junior inventory analyst at a UK online gift-ware wholesaler. The Operations Director: "Show me the health of our inventory, where the value and the risk sit, and let me test what a higher service level would cost us - in real time." 52

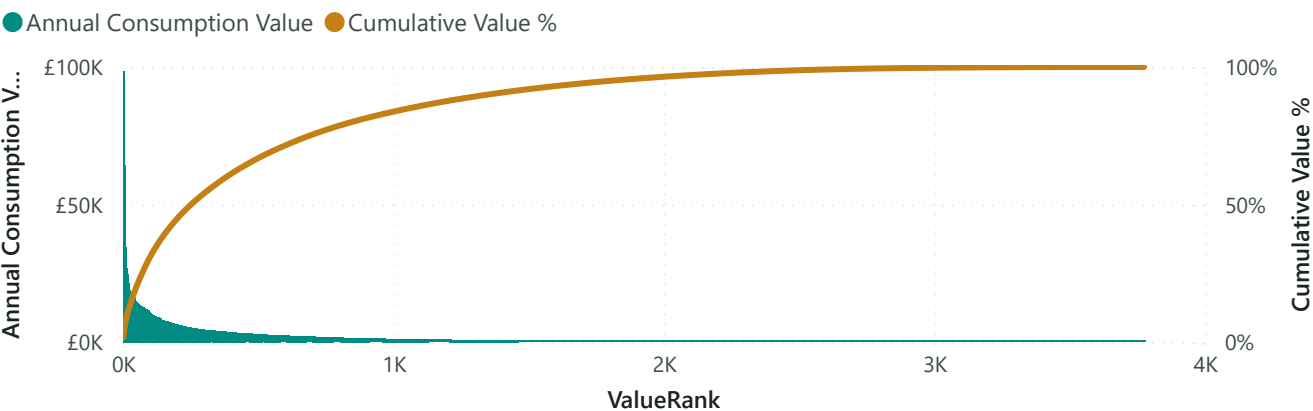
ASSUMPTIONS - Unit cost = median selling price x 0.60 (40% gross margin; source has no cost data). Lead time SIMULATED {2,4,6,8} weeks, deterministic from StockCode. On Hand SIMULATED from a code-derived weeks-of-cover rule - no random num...

Where is the value and how predictable is it?

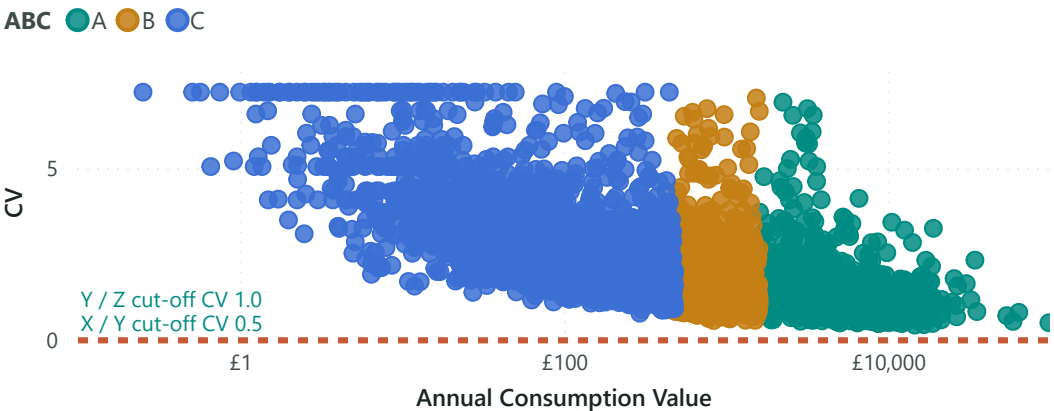
ABC x XYZ - SKU count and annual consumption value

XYZ	X		Y		Z
ABC	SKU Count	Annual Consumption Value	SKU Count	Annual Consumption Value	SKU Count
A	7	£91,994	254	£1,739,349	58
B			106	£103,502	85
C			13	£5,413	1,96
Total	7	£91,994	373	£1,848,265	3,39

Pareto - SKUs ranked by value, cumulative share on the line axis



Value vs predictability - X axis is log scale



Recommended inventory policy by segment

Segment	Profile	Recommended policy
AX	High value, steady demand	Tight automated continuous-review replenishment (ROP/EOQ). Frequent small or blanket POs. Review monthly.
AY	High value, variable demand	ROP with calculated safety stock; weekly planner review of forecast vs actual. Get
AZ	High value, lumpy demand	Do NOT rely on normal-distribution safety stock alone. Manual review, make-to-agreements for bulk orders. Highest exposure if import lead times lengthen.
BX	Medium value, steady	Automated periodic review, larger EOQ-based orders to cut ordering cost.
BY	Medium value, variable	Periodic review with calculated safety stock; check seasonality before each Q4 gi
BZ	Medium value, lumpy	Lower service target or order on demand; pool with similar items; check for pro

What does better service cost?

Service level

80.0%

99.0%

Lead time change (weeks)

-2

4

Service level mode

☐ Class-based (A 98% / B 95% / C 90%)

☐ Uniform (slider)

Ordering cost

☐ £20

☐ £30

Holding cost %

☐ 15%

☐ 20%

Safety stock investment

£679K

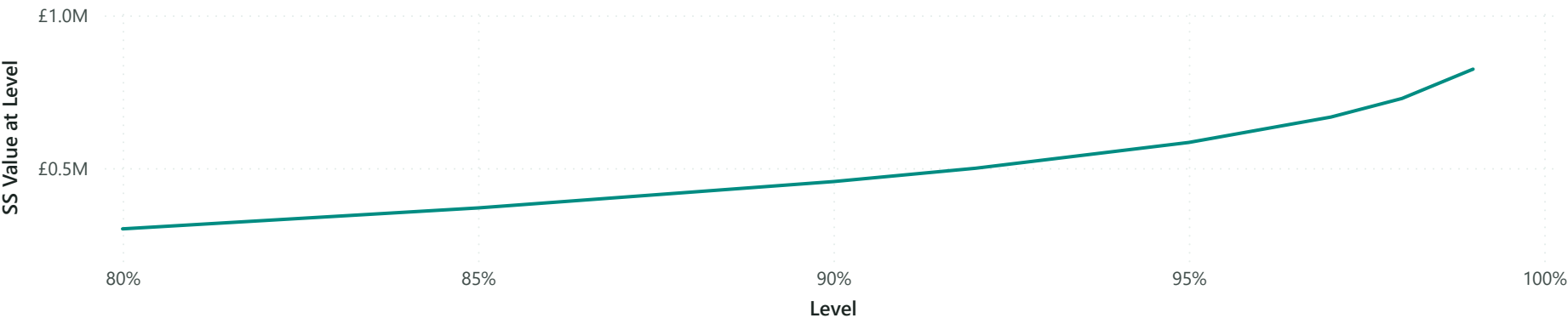
SKUs needing action

2,099

Working capital at ROP

£1M

Safety stock investment vs service level (uniform)



Service level

At a uniform 95.0% service level, safety stock is £583,623 - £0 more than at 95% (0.0%).

Supplier lead time

Lead times at plan. Move the slider to stress-test a supplier delay.

HOW TO READ THIS PAGE - Uniform mode applies one service level to every SKU. Class-based mode uses the live policy (A 98% / B 95% / C 90%) and deliberately ignores the service level slider, so set the mode to Uniform before demonstrating it. Safety stock grows with the SQUARE ROOT of lead time; reorder-point working capital grows LINEARLY with it, which is why the lead time slider moves the second card roughly three times as hard. That is the tariff and border-delay exposure - and unlike a service level, it is not a choice.

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What do we order today?

Below safety stock

1,308

Reorder now

791

Excess

222

Suggested order value

£1M

View

- ☐ Action
- ☐ Excess

ABC class

- ☐ A
- ☐ B

Planner action list

StockCode	Description	ABC_XYZ	Category	LeadTimeWeeks	On Hand (selected)	Safety Stock Units	Reorder Point Units	EOQ Units	Stock Status	Suggested Order Qty
10002	INFLATABLE POLITICAL GLOBE	CZ	Other	6	198	109	201	787	Reorder Now	790
10080	GROOVY CACTUS INFLATABLE	CZ	Other	2	23	22	34	720	Reorder Now	731
10123C	HEARTS WRAPPING TAPE	CZ	Other	8	0	2	3	64	Below Safety Stock	67
10135	COLOURING PENCILS BROWN TUBE	AZ	Other	8	501	395	729	1,076	Reorder Now	1,304
15030	FAN BLACK FRAME	CZ	Other	6	43	30	47	569	Reorder Now	573
15034	PAPER POCKET TRAVELING FAN	BZ	Other	6	253	1,026	1,787	5,601	Below Safety Stock	7,135
15058A	BLUE POLKADOT GARDEN PARASOL	BZ	Other	6	4	25	49	130	Below Safety Stock	175
15058B	PINK POLKADOT GARDEN PARASOL	BZ	Other	6	4	32	59	119	Below Safety Stock	174
15058C	ICE CREAM DESIGN GARDEN PARASOL	BZ	Other	6	5	46	74	122	Below Safety Stock	191
16010	FOLDING CAMPING SCISSOR W/KNIF & S	CZ	Other	6	3	6	9	316	Below Safety Stock	322
16011	ANIMAL STICKERS	CZ	Other	8	436	193	462	2,352	Reorder Now	2,378
16014	SMALL CHINESE STYLE SCISSOR	AZ	Other	6	3,314	3,336	4,866	4,587	Below Safety Stock	6,139

Total

HOW TO WORK THIS LIST - The View slicer switches the table between the SKUs that need ordering and the SKUs carrying excess; it is driven by a hidden [Action Filter] measure, so the row count here always matches the cards above. Suggested Order Qty is an order-up-to quantity: it brings stock to reorder point plus one EOQ, and is blank where no action is needed. Below Safety Stock is the more urgent of the two states - those SKUs have already spent

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Single SKU detail

SKU 10002 - INFLATABLE POLITICAL GLOBE | CZ | Other

Choose a SKU

10002

ABC class

C

XYZ class

Z

CV

2.28

Lead time (weeks)

6

Safety stock

109

Reorder point

201

EOQ

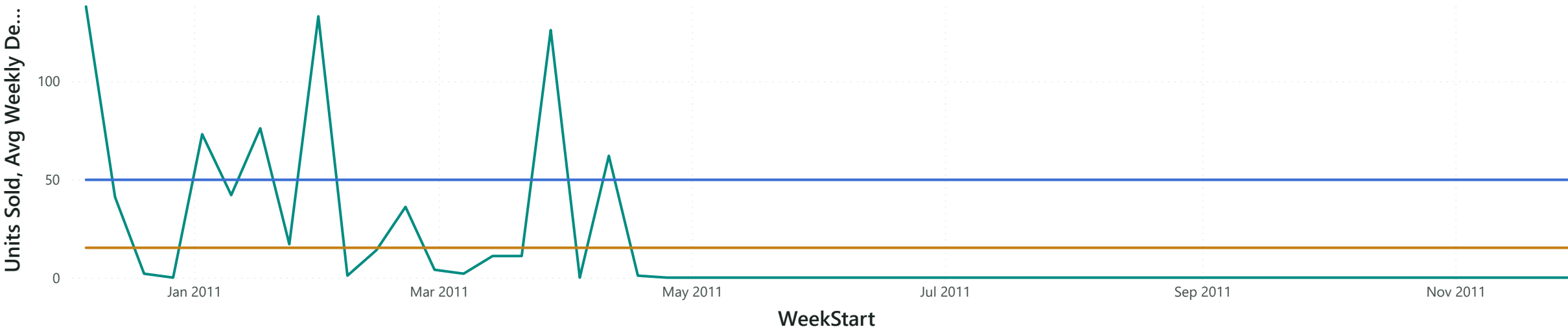
787

On hand

198

Weekly units, against the 52-week average and average + 1 standard deviation

Units Sold Avg Weekly Demand (all weeks) Avg Plus 1 SD (all weeks)



HOW TO READ THIS PAGE - The eight cards are single-SKU figures: they are blank unless exactly one SKU is selected in 'Choose a SKU'. The slicer is single-select, so picking a SKU replaces the previous one. Safety stock, reorder point and EOQ use the service level, lead time, ordering cost and holding cost set on page 3 - change a slider there and every card here moves with it. TO MAKE THIS A DRILL-THROUGH PAGE instead, drag Dim_SKU[StockCode] into

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Does this agree with the Excel model?

Verdict

MATCH ON POLICY - across all 500 shared SKUs the two models agree on XYZ class, EOQ, and, using the workbook's own ABC class, on safety stock and reorder point. The only residual is 1 SKU where the simulated on-hand snapshot lands exactly on a half unit and ...

Fact table shape

OK - 3775 SKUs x 52 weeks = 196300 rows

Simulated on hand

On hand differs on: 21908 (BI 911 vs XL 910)

SKUs shared	On hand differs	EOQ differs	XYZ differs	SS differs (raw)	ROP differs (raw)	ABC differs - scope	SS on Excel ABC	ROP on Excel ABC
500	1	0	0	210	210	210	0	0

SKU by SKU - Power BI beside the workbook

StockCode	Description	ABC	ABC (Excel)	XYZ	XYZ (Excel)	On Hand (selected)	On Hand (Excel)	Safety Stock Units	Safety Stock (Excel)	Safety Stock Diff	Safety Stock Units (Excel)
10002	INFLATABLE POLITICAL GLOBE	C		Z		198		109			
10080	GROOVY CACTUS INFLATABLE	C		Z		23		22			
10120	DOGGY RUBBER	C		Z		146		15			
10123C	HEARTS WRAPPING TAPE	C		Z		0		2			
10124A	SPOTS ON RED BOOKCOVER TAPE	C		Z		5		2			
10124G	ARMY CAMO BOOKCOVER TAPE	C		Z		4		2			
10125	MINI FUNKY DESIGN TAPES	B		Z		1,274		130			
10133	COLOURING PENCILS BROWN TUBE	B		Z		2,628		254			
Total											

Scope

☐ In Excel top-500

HOW THIS IS CHECKED - The workbook's SKU_Policy sheet is loaded as its own query and joined to Dim_SKU, so this compares against the real file, not against remembered numbers. Two of the cards are expected to be non-zero: ABC differs because Power BI ranks all 3,775 SKUs by cumulative consumption value while the workbook ranks only its own 500, so the 80/95% ...

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